Noah Hart

Gameplay Programmer | Technical Designer

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Summary

Gameplay programmer and technical designer with 6+ years of coding experience. As Lead Engineer on No Rail, I helped a 15-person team ship an Unreal Engine game on Steam. I build modular C++ gameplay, AI, and designer-friendly tools. I help teams deliver reliable features through debugging, clear documentation, and flexible systems that support iteration.

Technical Skills

Languages: C++, C#, C, Python, Lua, JavaScript, HTML; Unreal Blueprints
Engines & Tools: Unreal Engine 5, Unity, openFrameworks, Git, Perforce, Visual Studio, Blender, Maya, Jira, Trello
Gameplay & Architecture: GAS, AI (behavior trees, pathfinding, GOAP), state machines, procedural content generation (PCG), object pooling; Programming patterns (State, Command, Type Object)
Mathematics: Linear algebra (vectors, matrices, coordinate transforms, dot/cross products), calculus, trigonometry
Development: Software architecture, debugging, profiling, prototyping, problem solving, documentation, team leadership, collaboration, sprint-based production

Selected Projects

No Rail 2025–2026
Lead Engineer, Systems & Level Designer | UE5 / C++ | Team of 15 | Released on Steam
- Implemented modular train generation with streamed level instances, randomized car selection, and level-scoped spawners.
- Built interface-based survival, capture-zone, and defense objectives; managed station timing and return progression.
- Led engineering sprints, debugging, and integration with design and art to ship No Rail on Steam.

Squirrel Life In development
Solo Project | UE5 / C++
- Building reusable C++ minigames with shared base classes and Data Assets for designer tuning without code changes.
- Implemented obstacle pooling; use profiling to guide optimization and system documentation to support developer onboarding.

C++ Gameplay Systems & Combat Design 2025–2026
- Developed modular UE5 combat with GAS, health/damage, and stamina, separating shared systems from ability behavior.
- Built CineCaster, a multiplayer cinematic camera plugin with dynamic player switching.
- Built Pet Fish Simulator with seeded Perlin-noise terrain and constrained object placement using the Strategy pattern.

Graphics Engine Development 2025
- Extended an existing C++ graphics engine with abstraction interfaces that decouple platform-specific rendering APIs.
- Implemented matrix-based scaling, rotation, and translation; integrated physics, collision, input, UI, and event-driven audio in a Flappy Bird–style game.

AI Systems & Pathfinding 2026
- Implemented C++ A*, Dijkstra, steering, behavior trees, and GOAP; compared heuristics with runtime and search metrics.
- Combined A* navigation, ghost behavior trees, and procedural mazes in playable Pac-Man; deployed openFrameworks browser demos with WebAssembly.

Rockin Derby (Alt Control) & Rapid Prototyping 2024–2025
- Built three Unity/Unreal prototypes in 2–3 weeks each, including mobile games, under themed constraints, serious games and retro remixes.
- Built embedded ESP32/Arduino controllers for Alt Control, integrating custom hardware input with Unity.
- Processed accelerometer/gyroscope input for an IndieCade-nominated rocking-horse kart racer showcased at the university launch event.

Experience

Teaching Assistant — University of Utah 2025–2026
- As Head TA, mentored capstone teams delivering vertical slices; provided gameplay debugging and software architecture support.

Coding Tutor — University of California, Santa Cruz 2021–2023
- Tutored Python and C++ fundamentals, object-oriented programming, and algorithmic problem solving.

Education

University of Utah 2024–2026
Master of Entertainment Arts & Engineering (Engineering Track)
Simulated studio environment, sprint-based production, cross-disciplinary collaboration, and milestone-driven development

University of California, Santa Cruz 2019–2023
B.S. Computer Science | GPA: 3.8
Coursework: Data Structures, Computer Graphics, Artificial Intelligence, Game Design & Development